

KEERTHANA PM

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PROFILE

Cyber Security postgraduate with hands-on experience in enterprise security operations, automation, and security monitoring. Familiar with SIEM tools, security workflows, and infrastructure management, with exposure to various cybersecurity platforms and technologies. Passionate about building efficient security solutions and seeking opportunities in SOC, Security Operations, and Security Engineering.

EDUCATION

Global Academy of Technology 2020 – 2024
Bachelor of Engineering (Honours) - CGPA - 9.01 Bengaluru

UVCE 2024 – 2026
Master of Technology (Cyber Security) - CGPA - 8.88 Bengaluru

EXPERIENCE

Textron India Pvt Ltd Sept 2025 – Present
Student Intern – Enterprise Security Team Bengaluru

- Worked on **Splunk SIEM** for log ingestion and server class creation.
- Automated IIS log fixes, Splunk UF upgrades, and Tenable agent operations using **Ansible**.
- Gained hands-on exposure to **Databricks, XSOAR, Signal Sciences, Tenable, and CyberArk**.
- Utilized **CMDB** data to identify and analyze decommissioned servers within the enterprise environment.
- Currently working on feature enhancements for **SCDash 2.0**.
- Familiar with enterprise security workflows and **Microsoft Azure** basics.

Maitra Medtech Pvt Ltd (incubated in IISc) Mar 2024 – April 2024
Software developer intern Bengaluru

- Developed a React-based healthcare application with Bluetooth integration, data caching, and synchronization with MongoDB over WiFi.

A2Z Assist June 2023 – July 2023
Backend developer Bengaluru

- Managed PostgreSQL database operations and optimized backend performance for a dispatch portal project.

PROJECTS

Advanced Malware Detection using Hybrid Models | TensorFlow, LSTM, GRU, Streamlit

- Developed a hybrid **LSTM-GRU** based Android malware detection system achieving **93% accuracy** using over **250 static features** extracted from Android applications.
- Implemented both **static analysis** and **API-based testing** workflows with a **Streamlit** interface for real-time malware prediction and analysis.
- Integrated **XAI (Explainable AI)** techniques and orchestration methods to improve model interpretability, prediction transparency, and security analysis.

Digital Steganography | ReactJS, Tailwindcss, Cryptography

- Developed a secure image steganography system for hiding and extracting confidential messages within digital images using image processing techniques.
- Achieved **90%+ decoding accuracy** while maintaining minimal visual distortion in carrier images during encoding and decoding operations.

Network Intrusion Detection System | Python, Scikit-Learn, Pandas, Matplotlib

- Developed a machine learning-based intrusion detection system to classify network traffic into multiple attack categories including DOS, PROBE, R2L, U2R, and normal traffic.
- Implemented and evaluated multiple ML models including Random Forest, SVM, Decision Tree, AdaBoost, and MLP Classifier using Python and Scikit-Learn.

TECHNICAL SKILLS

Cybersecurity & Security Tools: Splunk (SIEM), Signal Sciences (WAF), Tenable (Vulnerability Management), CyberArk (PAM), Wireshark, Nmap, Autopsy, NetworkMiner, DumpIt, DiskDrill, ExifTool

Automation & DevOps: Ansible, Git, GitHub

Cloud Technologies: Microsoft Azure (Basics)

Programming & Web Technologies: Python, Java, C, HTML

Data Science & Machine Learning: NumPy, Pandas, Matplotlib, Scikit-Learn, TensorFlow, Keras, PyTorch, Databricks

Networking: OSI & TCP/IP Models, Ethernet, ARP, DHCP, ICMP, OSPF, RIP

Databases & API Tools: MySQL, MongoDB, PostgreSQL, Postman

CERTIFICATIONS

- Soft Computing a course certified by NPTEL
- Privacy and security in online social media certification from NPTEL
- Splunk Search Expert

PUBLICATION

- **Keerthana PM** , “*Comparative Study of Various Optimizers and Transfer Learning Methods of Rice Leaf Disease Detection using CNN*”. Published paper on the comparative study of optimizers and transfer learning methods for rice leaf disease detection using Convolutional Neural Networks (CNNs). The study explores the effectiveness of different optimizers and transfer learning techniques in enhancing the accuracy and efficiency of disease detection models. Results provide valuable insights for improving agricultural disease detection systems, contributing to advancements in precision agriculture and food security.
Link: <https://ieeexplore.ieee.org/document/10420090>

ACTIVITIES

- **National Cadet Corps (2021–2024):** Actively participated in various training programs and community service initiatives as a dedicated NCC cadet. Held the position of *Lance Corporal*, demonstrating leadership and discipline in all assigned responsibilities.
- **NSS Volunteer:** Contributed to social awareness programs, community service, and outreach initiatives as part of the National Service Scheme (NSS), fostering civic responsibility and teamwork.